

Escola Tècnica Superior d'Enginyeria Informàtica

Final Technical Report

Model-Agnostic Interpretability in Predictive Systems

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Abstract

This report applies model-agnostic XAI methods to interpret Random Forest models for bike rental demand and housing prices. By utilizing 1D and 2D Partial Dependence Plots (PDP), we isolate marginal effects such as non-linear climatological impacts and the primary influence of living area on valuation. This approach effectively transforms "black-box" models into transparent tools for informed decision-making.

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1. Introduction

Modern predictive analytics often rely on "black-box" models like Random Forests to achieve high accuracy. However, sometimes, especially for business, accuracy is not enough to make strategic decisions if there is a lack of explanation. Not only the predictive capacity is valuable, but its interpretability: the ability to explain the impact of variables like climate conditions or geographic location.

With this in mind, this report evaluates housing price dynamics and bike rental demand performing an analysis based on Partial Dependence Plots to move beyond simple predictions and understand the logic behind the results. This approach turns the "black-box" problems into a clear tool for making more informed decisions.

2. Bike Rental Demand Analysis

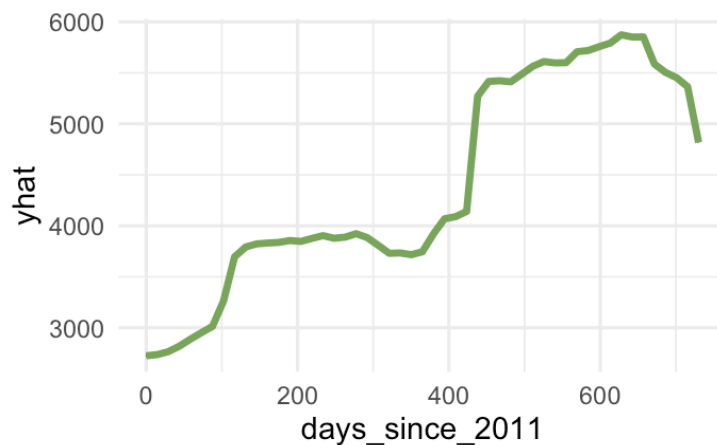


Figure 1: PDP of days_since_2011

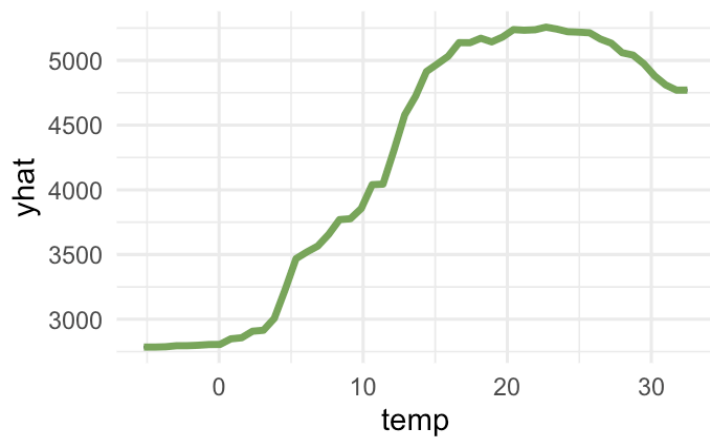


Figure 2: PDP of temperature

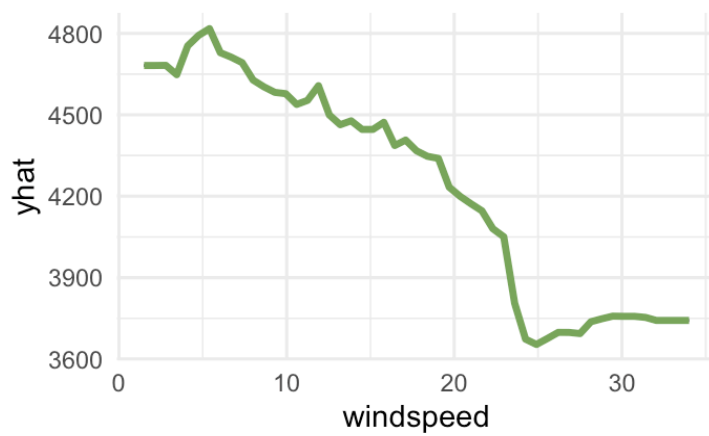


Figure 3: PDP of windspeed

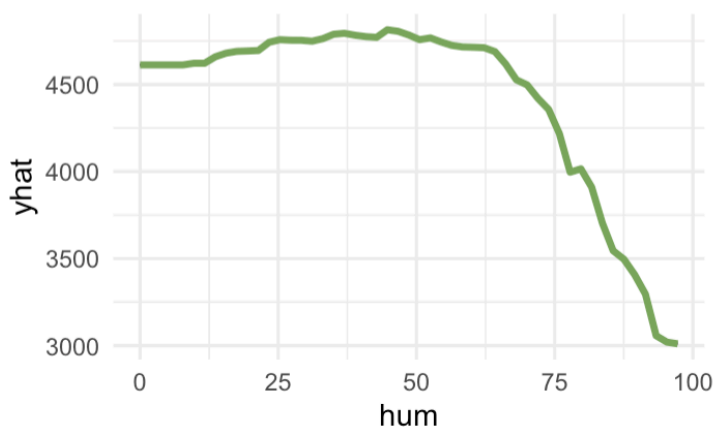


Figure 4: PDP of humidity

The weather situation variable (weathersit) was also preprocessed by mapping the original numeric codes to descriptive categories (“GOOD”, “MISTY”, “RAIN”, “HEAVY RAIN”), so that any future models using this feature remain interpretable.

Partial Dependence Analysis allows the isolation of the effects of critical variables on rental volume predictions, revealing how the Random Forest model captures the business logic:

- **Days since 2011 (Figure 1):** The plot shows an upward growth trend in demand. Two significant slopes are observed, one around 100 days and another near 420 days, reflecting the service's consolidation.
- **Temperature (Figure 2):** There is a clear positive relationship between temperature and bike usage. Demand surges starting at 5°C and reaches its peak around 25°C, after which a slight drop is observed, indicating that extreme heat discourages cycling.
- **Wind Speed (Figure 3):** Wind speed has a near-linear negative influence. As wind increases, demand decreases, with a particularly sharp drop once it exceeds 20 km/h. It appears that wind is a decisive factor for users.
- **Humidity (Figure 4):** Humidity maintains a neutral to slightly positive effect until it exceeds 65%, at which point rental predictions drop drastically. This suggests the model identifies high humidity levels as a proxy for a high probability of precipitation.

4. Climatological Interactions

To analyze the interaction between multiple variables and their combined effect on bike rental demand, a **Bidimensional Partial Dependence Plot** was generated. Due to the computational intensity of calculating predictions across a 2D grid, a set of random samples was extracted from the dataset to generate the PDP data (Figure 5).

The 2D plot reveals that the highest demand occurs when temperatures are warm (20-25°C) and humidity is relatively low (below 50%). It seems like high humidity levels drastically reduce predicted rentals even when the temperatures are at their best.

Regarding the marginal density plots, the top curve of the temperature indicates that temperature data is most concentrated between 15°C and 25°C, which aligns with rental's prediction peak. The curve on the right illustrates the distribution of humidity levels, with a

peak suggesting that most observations fall between 50% and 75%. To summarize, marginal plots provide a context by showing where the model's predictions are supported by the highest density of real-world data, ensuring that interpretation align with the most frequent weather patterns experienced in the dataset.

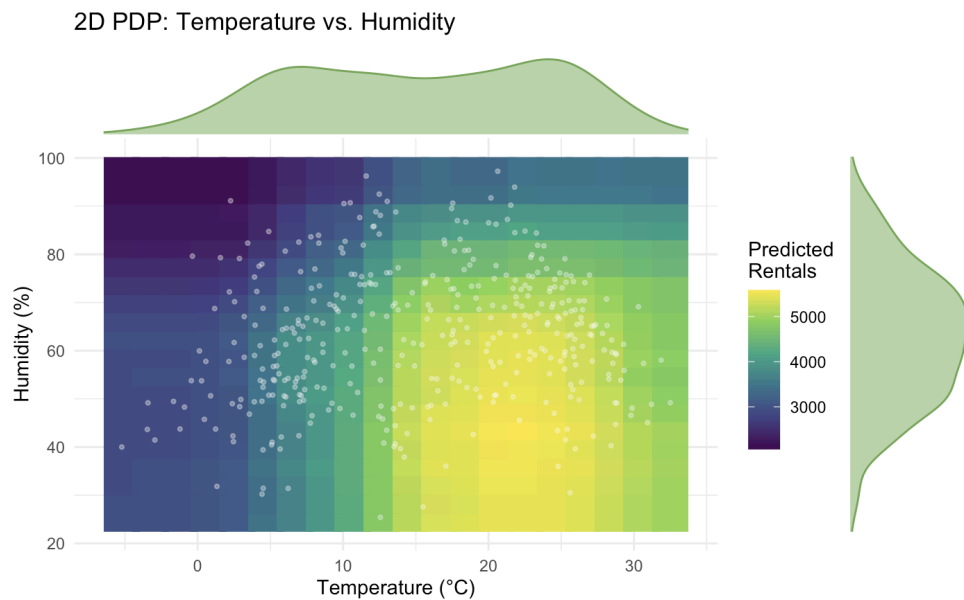


Figure 5: 2D PDP of Humidity vs. Temperature

5. Housing Market Valuation

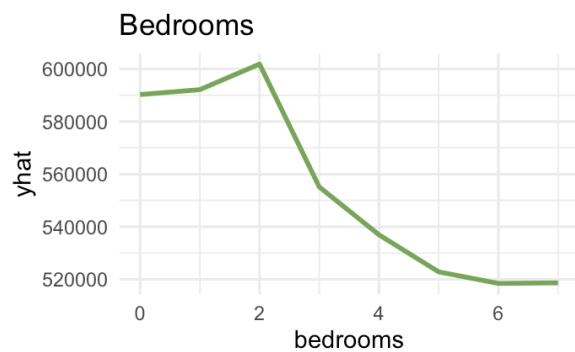


Figure 6: PDP of Bedrooms

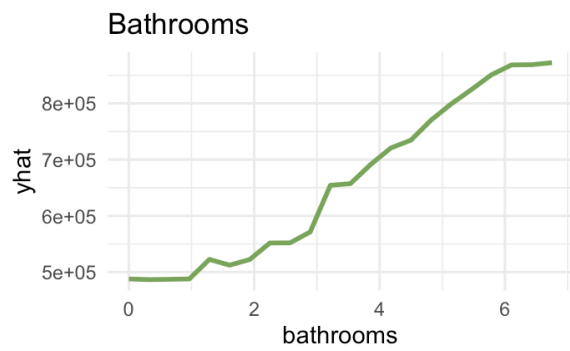


Figure 7: PDP of Bathrooms

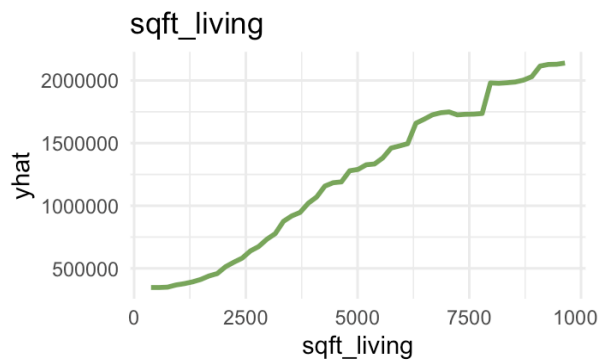


Figure 8: PDP of sqft_living

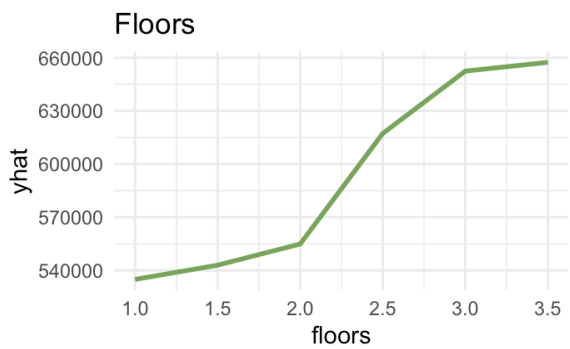


Figure 9: PDP of floors

To analyse the factors influencing housing prices, a random forest model was fitted using the `kc_house_data` dataset. The target variable is the sale price, and the explanatory variables are the number of bedrooms and bathrooms, the living area in square feet (`sqft_living`), the lot

size (sqft_lot), the number of floors, and the year the home was built. To reduce computational cost, the model was trained on a random sample of the available records.

The Partial Dependence Plots show a clearly increasing, but not perfectly linear, relationship between sqft_living and the predicted price: as living area grows, the expected price rises, especially for medium and large houses. This confirms that living space is one of the most influential factors for the model. For bedrooms and bathrooms, the PDPs also display positive effects, but with a step-wise pattern: moving from very few to a reasonable number of rooms increases the predicted price, while adding extra bedrooms beyond a certain point leads to much smaller gains.

In contrast, the influence of floors is moderate and very small in some ranges, suggesting that the number of storeys is less relevant than size or comfort features. Overall, the PDP analysis indicates that the random forest bases its decisions mainly on the usable living area and the basic equipment of the property (especially bathrooms), while other aspects, such as lot size or the exact number of floors, play a secondary role in determining the predicted price.

6. Conclusions

This study demonstrates the capacity of model-agnostic methods in translating complex algorithmic outputs into actionable business intelligence. Through the application of Partial Dependence Plots, we successfully decoded the "black-box" nature of Random Forest models for two distinct sectors. For bike rentals, the analysis proved that while there is an upward growth trend, environmental constraints such as high humidity (above 65%) and extreme wind speeds (exceeding 20 km/h) act as critical limiting factors. The 2D interaction analysis further revealed that the business reaches its peak performance under warm, dry conditions, typically between 20-25°C with low humidity.

In the housing market valuation, the results confirmed that functional living space is the primary driver of price. The models reveal a sharp, nearly linear relationship between square footage and value, whereas secondary features like the number of floors or bathrooms show diminishing marginal returns after a certain threshold. Ultimately, integrating XAI techniques

ensures that predictive models are not only accurate but also transparent and trustworthy, turning complex data patterns into a clear tool for informed strategic decision-making.